



CHEMISTRY 1

ACADEMIC

Grade 11/12

Curriculum Guide

CHEMISTRY 1
Prerequisite: None

Course Description:

This course introduces foundational concepts in chemistry, focusing on the structure, properties, and interactions of matter. The learners develop skills in accurate measurement and laboratory practices. They explain the development of the quantum mechanical model, write electronic configurations, and analyze periodic trends to predict the properties of elements.

The course also explores how chemical bonding, molecular geometry, polarity, and intermolecular forces collectively influence the properties and behavior of substances. The learners use these concepts to explain chemical phenomena and make predictions based on scientific models.

Throughout the course, the learners apply chemistry concepts to environmental science, community-based issues, and industrial processes, with emphasis on Green Chemistry and the United Nations Sustainable Development Goals. The course promotes scientific inquiry, systems of thinking, evidence-based reasoning, and the responsible application of chemistry in addressing societal and environmental challenges.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Atomic Structure and Periodicity		
1. Introduction to Chemistry	1. chemistry plays an essential role in solving real-world problems and contributes to various industries and career pathways;	1. describe how chemistry contributes to solving real-world problems in health, environment, and industry as well as career opportunities in chemistry-related fields;
2. Properties of Matter	2. matter has observable and measurable properties that can be used to classify, describe, and explain changes in substances;	2. describe the physical and chemical properties of matter, and explain how these properties are used in real-life applications and in identifying substances;
3. Measurement and Laboratory Skills	3. accurate measurement, precision, and proper use of scientific notation and significant figures are essential for reliable experimental results; 4. experimental results are affected by sources of error which can be minimized;	3. explain the importance of accuracy and precision in assessing the quality of experimental results;
		4. identify sources of experimental error and suggest ways to improve the reliability of experimental results;
		5. apply significant figures and scientific notation in calculations and unit conversions using accurate measurements obtained from basic laboratory equipment (e.g., graduated cylinder, balance, ruler, thermometer);
4. The Quantum Theory and the electronic structure of atoms	5. the contributions of Max Planck, Niels Bohr, and Erwin Schrödinger helped in the development of Quantum Theory, which explains the behavior of subatomic particle;	6. describe the development of quantum mechanical model;
		7. write the electronic configurations based on quantum mechanical principles;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	6. electrons of elements are arranged in energy levels, subshells, and orbitals based on the quantum mechanical model;	8. predict main energy levels, sublevels, and orbitals in relation to the electronic configuration of elements;
5. Periodicity	7. periodic trends arise from variations in electron configuration and effective nuclear charge, enabling the prediction of properties; and	9. explain how Dmitri Mendeleev developed the periodic table of elements;
		10.use periodic trends to predict the properties of elements, including atomic radius, ionization energy, electron affinity, and electronegativity;
6. Isotopes	8. isotopes of elements can have both beneficial and harmful effects.	11.identify natural and human-made radioisotopes and their properties in terms of the type of radiation emitted and their half- life; and
		12.describe the applications of isotopes in medicine, industry, and environmental science.
Performance Standards <i>The learners can apply science process skills to obtain accurate and reliable measurements and recognize sources of error. They also apply significant figures and scientific notation in calculations. They can explain the development of quantum mechanical models, write electron configurations, and use periodic trends to predict the properties of elements and describe their applications. The learners can also explain what isotopes are and how they may be useful and possibly harmful.</i>		
Suggested Performance Tasks - Conduct a laboratory investigation to measure a physical quantity (e.g., density) and record data using correct units, significant figures, and scientific notation. - Make a two-minute video presentation explaining how radioactive isotopes are used in the treatment of cancer and goiter.		
Chemical Bonding and Intermolecular Forces		

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
7. Chemical Bonding and Compounds	9. chemical bonding involves the transfer or sharing of electrons and determines the structure and properties of substances;	13. identify ionic and covalent compounds found in everyday life, environment, and industries;
		14. explain that chemical bonds may be ionic or covalent and differentiate them in terms of the electrons involved;
		15. name and write formulas of ionic and covalent compounds using International Union of Pure and Applied Chemistry (IUPAC) nomenclature;
		16. draw Lewis structures of ionic and covalent compounds by applying the Octet Rule;
		17. conduct experiments to differentiate the properties of ionic and covalent compounds, such as solubility, melting and boiling points, and electrical conductivity;
8. Electronegativity and Polarity	10. differences in electronegativity influence bond polarity;	18. explain how electronegativity differences determine bond polarity;
9. Molecular Geometry and Polarity	11. molecular geometry determines the shape and polarity of molecules; and	19. predict molecular polarity based on VSEPR theory and molecular geometry;
10. Intermolecular Forces and Properties of Substances	12. intermolecular forces affect the physical properties and behavior of substances.	20. use diagrams to differentiate the intermolecular forces from intramolecular forces;
		21. describe different types of intermolecular forces, such as London dispersion forces, dipole-dipole forces, and hydrogen bonding;
		22. explain the effects of intermolecular forces on liquid properties such as surface tension, vapor pressure, boiling point, and melting point;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
		23.conduct an investigation comparing substances based on intermolecular forces and their physical properties; and
		24.analyze how chemical bonding, molecular polarity, and intermolecular forces influence product development and the design of sustainable solutions.
Performance Standards <i>The learners</i> can write names and formulas particularly for compounds commonly used in everyday life, industries, and environment. They can differentiate between ionic and covalent bonds and use diagrams to write Lewis structures of ionic and covalent compounds while applying the Octet Rule. They can also conduct experiments to differentiate the properties of ionic and covalent compounds. The learners predict polarity based on electronegativity and molecular geometry. They can explain how intermolecular forces influence the physical properties of substances and conduct an investigation to compare substances based on these interactions.		
Suggested Performance Tasks - Conduct experiments to investigate the physical properties of ionic and covalent compounds, such as solubility, melting and boiling, and conductivity. - Conduct an investigation comparing substances (e.g., water, ethanol, and acetone) based on physical properties such as boiling point or solubility and explain the results using intermolecular forces.		